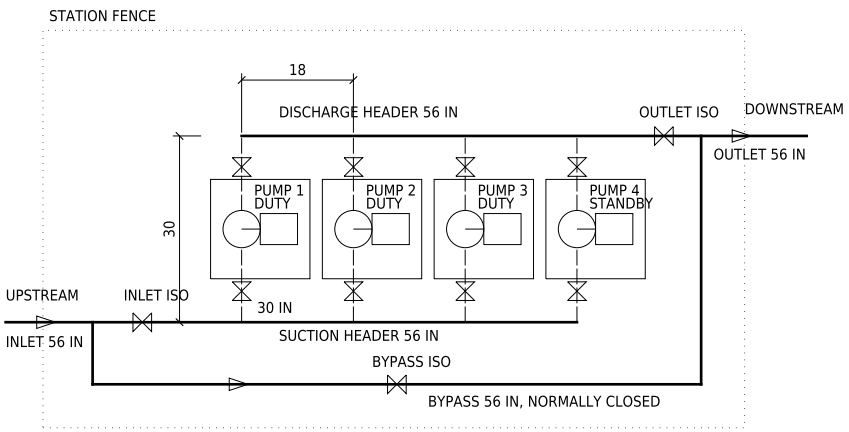


LEGEND

- 56 IN PIPING
- 30 IN PIPING
- ISOLATION VALVE
- CENTRIFUGAL PUMP AND MOTOR
- FLOW DIRECTION
- STATION FENCE
- PUMP PAD

NOTES

1. PLAN VIEW. NOT A P&ID.
2. SINGLE-LINE PIPING, SIZES AS LABELLED.
3. 3 DUTY PUMPS + 1 STANDBY, IN PARALLEL.
4. HYDRAULIC INPUTS ARE ASSUMED.
5. NOT CHECKED: NPSH, MAOP, SURGE, DRA.
6. GENERIC STATION, NOT A REAL FACILITY.
7. UNITS: METRES. A3 AT 1:500.

N

SCALE 1:500

0 20 40 M

DESIGN BASIS (ASSUMED)		
FLOW Q	2.5	MMBPD
FLOW Q	4.600	M3/S
CRUDE GRAVITY	33	DEG API
KINEMATIC VISCOSITY	8	CST
PIPE ROUGHNESS	0.046	MM
STATION SPACING L	100	KM
SUCTION PRESSURE, NEXT STN	8	BAR(G)
STATION INTERNAL LOSS	1.0	BAR
PIPE OD X WALL	56 X 0.625	IN
PIPE ID (USED FOR V)	1.3906	M

PUMP DUTY AND POWER		
PUMP DIFFERENTIAL PRESSURE	39.2	BAR
DIFFERENTIAL HEAD	466	M
DISCHARGE PRESSURE	46.2	BAR(G)
FLOW PER DUTY PUMP	1.533	M3/S
FLOW PER DUTY PUMP	5,520	M3/H
LATERAL VELOCITY (ID 0.737 M)	3.60	M/S
HYDRAULIC POWER	18.1	MW
SHAFT POWER, EFF 0.85	21.2	MW
SHAFT POWER PER PUMP	7.1	MW
MOTOR INPUT, EFF 0.96	22.1	MW

FLUID AND LINE		
SPECIFIC GRAVITY	0.8602	-
DENSITY	859.3	KG/M3
VELOCITY	3.03	M/S
REYNOLDS NUMBER	526,000	-
FRICTION FACTOR	0.0135	-
LINE PRESSURE LOSS	38.2	BAR

FORMULAS
SG = 141.5 / (API + 131.5)
RHO = SG X 999.016 KG/M3
V = 4 Q / (PI D D) RE = V D / NU
F = SWAMEE-JAIN (EPS/D, RE)
DP = F (L/D) (RHO V V / 2) + STATION LOSS
H = DP / (RHO G) P = Q DP
SHAFT = P / EFF_P MOTOR = SHAFT / EFF_M

BOOSTER PUMP STATION

PLOT PLAN AND HYDRAULIC BALANCE

PROJECT: HORMUZ BYPASS STUDY
SHEET 1 OF 1 REV A

SCALE 1:500 (A3) UNITS: M
DRAWN ___ CHK ___ DATE 2026-10-13